Aim:
9. Set up Apache Spark locally and count the frequency of words in a text file.

Input.txt:
Shreyash S091
Apache Spark is a powerful data processing framework.
Spark can process large amounts of data.
Python is easy to learn.
Spark and Python are useful for data processing.

Code:
```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("WordCount") \
    .master("local[*]") \
    .getOrCreate()

text_file = spark.sparkContext.textFile("input.txt")

words = text_file.flatMap(lambda line: line.split())

word_pairs = words.map(lambda word: (word, 1))

word_counts = word_pairs.reduceByKey(lambda a, b: a + b)

print("\nShryeash Kadam S091")
print("\nWord Frequency:")


for word, count in word_counts.collect():
    print(word, ":", count)

spark.stop()
```

Output:

```
Shryeash Kadam S091

Word Frequency:
C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Pract
C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Pract
C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Pract
C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Pract
Shreyash : 1
S091 : 1
powerful : 1
framework. : 1
large : 1
of : 1
data. : 1
Python : 2
easy : 1
to : 1
learn. : 1
and : 1
are : 1
useful : 1
for : 1
Apache : 1
Spark : 3
is : 2
a : 1
data : 2
processing : 1
can : 1
process : 1
amounts : 1
processing. : 1

Process finished with exit code 0
```

Aim:
10. Filter rows in a CSV file using Spark DataFrames where a numeric column exceeds a threshold.

Stock.csv:
```
date,symbol,open,high,low,close,volume
2017-01-03,AAL,44.50,46.20,44.10,45.80,1250000
2017-01-04,AAL,45.90,47.10,45.50,46.75,1320000
2017-01-05,AAL,46.80,48.20,46.30,47.90,1450000
2017-01-06,AAL,47.95,49.00,47.50,48.60,1380000
2017-01-09,AAL,48.70,50.20,48.20,49.85,1510000
```

Code:
```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("FilterRows") \
    .master("local[*]") \
    .getOrCreate()

data = spark.read.csv("stocks.csv", header=True, inferSchema=True)

threshold = 40.0

filtered_data = data.filter(data["close"] > threshold)

print("\nRows where Close Price > 40:")
filtered_data.show()

print("Total Rows:", filtered_data.count())

spark.stop()
```

Output:

```
Rows where Close Price > 40:
+----------+------+-----+----+----+-----+-------+
|      date|symbol| open|high| low|close| volume|
+----------+------+-----+----+----+-----+-------+
|2017-01-03|   AAL| 44.5|46.2|44.1| 45.8|1250000|
|2017-01-04|   AAL| 45.9|47.1|45.5|46.75|1320000|
|2017-01-05|   AAL| 46.8|48.2|46.3| 47.9|1450000|
|2017-01-06|   AAL|47.95|49.0|47.5| 48.6|1380000|
|2017-01-09|   AAL| 48.7|50.2|48.2|49.85|1510000|
+----------+------+-----+----+----+-----+-------+

Total Rows: 5
```